

Entegris, Inc.

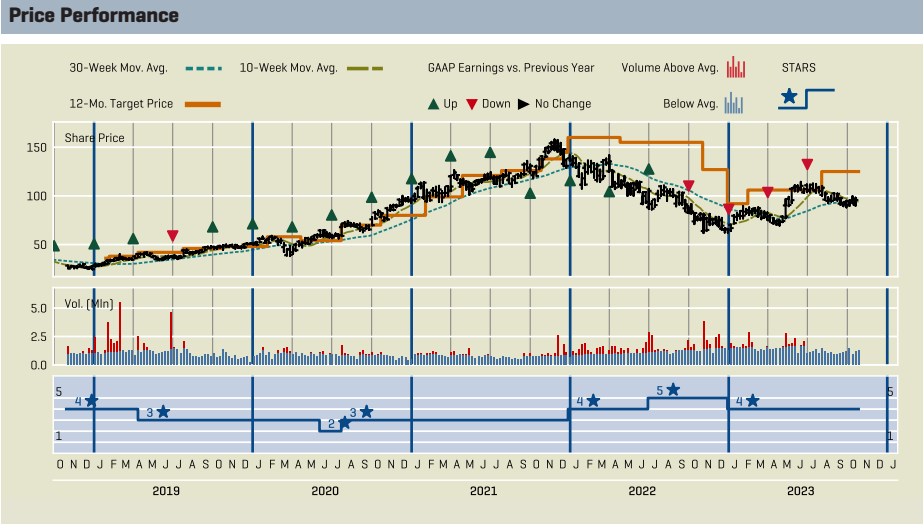
Recommendation **BUY** ★★★★★

Price USD 87.82 [as of market close Oct 27, 2023] **12-Mo. Target Price** USD 125.00 **Report Currency** USD **Investment Style** Mid-Cap Growth

Equity Analyst Brooks Idlet, CFA

GICS Sector Information Technology **Sub-Industry** Semiconductor Materials and Equipment **Summary** Entegris provides specialty chemical and material integrity solutions for microelectronics and life science applications.

Key Stock Statistics [Source: CFRA, S&P Global Market Intelligence (SPGMI), Company Reports]							
52-Wk Range	USD 114.47 - 61.75	Oper.EPS2023E	USD 2.64	Market Capitalization[B]	USD 13.18	Beta	1.26
Trailing 12-Month EPS	USD 2.99	Oper.EPS2024E	USD 3.71	Yield [%]	0.46	3-yr Proj. EPS CAGR[%]	8
Trailing 12-Month P/E	29.37	P/E on Oper.EPS2023E	33.27	Dividend Rate/Share	USD 0.4	SPGMI's Quality Ranking	B+
USD 10K Invested 5 Yrs Ago	36,260.0	Common Shares Outstg.[M]	150.00	Trailing 12-Month Dividend	USD 0.5	Institutional Ownership [%]	100.0



Source: CFRA, S&P Global Market Intelligence
Past performance is not an indication of future performance and should not be relied upon as such.
Analysis prepared by **Brooks Idlet, CFA** on Sep 18, 2023 03:56 PM ET, when the stock traded at **USD 92.96**.

Highlights

- ▶ We project revenue growth of 5% in 2024 after our view of an 11% rise in 2023 (+43% in 2022). Consumer-led cyclical softness has made 2023 challenging, but we think ENTG will outperform other WFE participants given its strong position in advanced technologies, evidenced in Q2 via the resiliency of its Microcontamination Control segment [revenue up 3% Y/Y], which includes sub-5nm filtration products, and relative strength in its SCEM segment (-11% Y/Y) via its advanced deposition solutions. Sales growth should accelerate in 2024 as demand for complex chip architectures and node transitions outweighs a soft macro, with additional growth provided from the continued integration of recently-acquired CCMP.
- ▶ We project gross margin will expand to 44% in 2024 from 43% in 2023 [flat with 2022]. ENTG is investing in capacity expansions, and costs associated with the ramp of its new Taiwan facility should turn to benefits as initial shipments begin later in 2023.
- ▶ The CCMP acquisition pushed long-term debt to \$5.5B, with deleveraging now a top priority. ENTG's gross leverage ratio is 5.2x [3.5x targeted by end-2024]. We project FCF improves to \$345M in 2024 from \$230M in 2023, driven by sales growth.

Investment Rationale/Risk

- ▶ Our Buy rating reflects our view that ENTG shares are undervalued. We see ENTG's revenue as highly durable given [1] the critical nature of its solutions in enabling customers' technology transitions, the spending for which is more resilient than market-based capacity expansions; [2] the largely consumable nature of its solutions resulting in ~80% recurring revenue; and [3] the breadth of its products, with no product representing over 4% of sales in 2022. We like the ~\$6B acquisition of CMC Materials [CCMP], which increased combined entity recurring revenue (from 70%) and provides more end-to-end solutions for customers, although we note the associated higher leverage may give investors pause. We expect ENTG to benefit from positive long-term trends in the semiconductor industry, such as AI and electrification.
- ▶ Primary risks include an inability to deleverage via non-core asset sales, especially in a higher-for-longer rate environment, and the risk of global share loss as countries prioritize supply chain regionalization.
- ▶ Our target price of \$125, 33.7x our 2024 EPS view, is slightly above semi-cap peers based on ENTG's history of outpacing WFE and its growing exposure to recurring revenue.

Analyst's Risk Assessment

LOW **MEDIUM** **HIGH**

Our risk assessment reflects the competitive nature of the company's business and high degree of cyclicality and capital intensity associated with the semiconductor industry. The company's predominantly consumables-based business, which is well-balanced across logic and memory markets, helps absorb business volatility during uncertain times and counter associated risks, in our opinion.

Revenue/Earnings Data

Revenue (Million USD)					
	1Q	2Q	3Q	4Q	Year
2024	E 892	E 927	E 977	E 1,011	E 3,807
2023	922	901	E 902	E 909	E 3,634
2022	650	692	994	946	3,282
2021	513	571	579	635	2,299
2020	412	448	481	518	1,859
2019	391	379	394	427	1,591

Earnings Per Share (USD)

	1Q	2Q	3Q	4Q	Year
2024	E 0.77	E 0.89	E 0.98	E 1.07	E 3.71
2023	0.65	0.66	E 0.64	E 0.69	E 2.64
2022	1.06	1.00	0.85	0.83	3.73
2021	0.70	0.85	0.92	0.96	3.44
2020	0.55	0.60	0.67	0.71	2.54
2019	0.50	0.39	0.50	0.55	1.93

Fiscal Year ended Dec 31. EPS Estimates based on CFRA's Operating Earnings; historical earnings are adjusted. In periods where a different currency has been reported, this has been adjusted to match the current quoted currency.

Dividend Data

Amount [USD]	Date Decl.	Ex-Div. Date	Stk. of Record	Payment Date
0.1000	Oct 18	Oct 31	Nov 01	Nov 22 '23
0.1000	Jul 19	Aug 01	Aug 02	Aug 23 '23
0.1000	Apr 19	May 02	May 03	May 24 '23
0.1000	Jan 18	Jan 31	Feb 01	Feb 22 '23

Dividends have been paid since 2017 . Source: Company reports
Past performance is not an indication of future performance and should not be relied as such.
Forecasts are not a reliable indicator of future performance. Dividends paid in currencies other than the Trading currency have been accordingly converted for display purposes.

Entegris, Inc.

Business Summary Sep 18, 2023

CORPORATE OVERVIEW: Entegris [ENTG] develops, manufactures, and supplies products and materials used in processing and manufacturing in the semiconductor and other high-tech industries. For the semiconductor industry, the company's products maintain the integrity of critical materials. Its products are used during many key steps in the semiconductor manufacturing process, including deposition, etch, resist stripping, cleaning, lithography, ion implantation, and planarization, and it also provides products used in wafer and reticle transport, chemicals handling, as well as wafer and package testing. For other applications, its products and materials are used to manufacture items like flat panel displays, chemicals, solar cells, gas lasers, and critical components for aerospace, glass manufacturing, and biomedical applications.

The company offers both unit-driven and capital-driven products. ENTG's unit-driven and consumable product class includes liquid filters, metal-based gas filters, gas purifiers, wafer shippers, disk-shipping containers, and test assembly and packaging products. It also makes consumable graphite and silicon carbide components used in plasma etch, ion implant, and chemical vapor deposition processes in semiconductor manufacturing. The company's capital expense-driven products include its components, systems, and subsystems that permit semiconductor and other electronics manufacturers to monitor and control the flow and condition of process liquids used in these manufacturing processes, as well as its process carriers that protect the integrity of in-process wafers.

The company reports its financial performance based on four reporting segments: Specialty Chemicals and Engineered Materials [SCEM], Microcontamination Control [MC], Advanced Materials Handling [AMH], and Advanced Planarization Solutions [APS]. Starting in Q3 2023, the SCEM and APS divisions will be reported on a combined basis. The SCEM segment [22% of 2022 sales] provides high-performance and high-purity process chemistries, gases, and materials that enable enhanced device performance. Delivering advanced materials at high purity is critical to enabling the performance of leading-edge logic and memory applications. The MC segment [28%] offers solutions to purify critical liquid chemistries and gases used in semiconductor manufacturing processes and other high-tech industries. The AMH segment [21%] develops solutions to monitor, protect, transport, and deliver critical liquid chemistries, wafers, and substrates for a broad set of applications. These systems and products improve customer yields by protecting wafers from abrasion, degradation, and contamination during manufacturing and transportation. Within this segment, ENTG leads the market for 300mm wafer front-opening unified pods [FOUPs] and 200mm wafer standard mechanical interface pods [SMIFs]. The APS segment [29%], a result of ENTG's 2022 acquisition of CMC Materials, includes an end-to-end suite of chemical mechanical planarization [CMP] solutions, including CMP slurries, CMP pads, and formulated cleans and other electronic chemicals to remove excess material deposited during semiconductor manufacturing and level/smooth the surfaces of semiconductor device layers.

IMPACT OF MAJOR DEVELOPMENTS: On July 6, 2022, ENTG closed its acquisition of CMC Materials, a company that develops and manufactures CMP consumable products, for a total purchase price of \$6.0B, including \$3.8B in cash. The deal gives ENTG greater scale and makes its product portfolio more comprehensive, allowing customers to look to ENTG for more end-to-end solutions. The company issued a sizeable \$2.495B in floating-rate [SOFR + 3%] debt to finance the acquisition, and in 2023, it has made non-core asset sales a priority in order to reduce its debt load. In March, ENTG announced the sale of its QED business [in its SCEM segment] for \$135M; in May, it announced the sale of its Electronic Chemicals business [in its APS segment] for \$700M; and in June, it announced the termination of its distribution agreement for copper plating chemistries with Element Solutions in exchange for \$200M.

MARKET PROFILE: The company's major semiconductor customer groups include semiconductor manufacturers, semiconductor capital equipment manufacturers, gas and chemical manufacturing companies, and manufacturers of high-precision electronics. It has a relatively diverse customer base, with its top 10 customers accounting for 43% of sales in 2022, led by TSMC at 12%. ENTG has a very broad product portfolio, consisting of ~30,000 products, with no product representing over 4% of sales in 2022. The company works with major advanced technology enablers, providing advanced technology, such as sub-5nm process filtration products, advanced deposition materials for next-gen transistor and interconnect technologies, CMP solutions for advanced memory applications, advanced reticle pods for EUV applications, and advanced 300mm wafer carriers. Sales by geography are not overly concentrated in a single region, with 2022 sales led by customers in North America [24%], followed by Taiwan [20%], China [15%], South Korea [13%], Japan [11%], Europe [10%], and Southeast Asia [7%].

FINANCIAL TRENDS: Semiconductor equipment manufacturers have historically been highly cyclical, with periods of strong growth and high margins eventually leading to over-investment and excess supply followed by periods of weakness, with the cycle repeating when demand eventually catches up to and outstrips supply. Sales of unit-driven products [recurring revenue] represented 80% of total sales in 2022. Total 2022 sales of \$3.28B were up 42.8% Y/Y from \$2.30B in 2021, with \$594M of organic growth driven by strong demand for wafer and fluid handling and \$496M of growth added via the CMC acquisition. As of July 1, 2023, ENTG had cash and short-term investments of \$600M and long-term debt of \$5.49B. Debt to trailing-12-month EBITDA was 5.20x at July 1, 2023, compared to 5.68x at year-end 2022. The company hedges its \$2.495B in floating rate debt with an interest rate swap [expiring December 30, 2025]. No material debt maturities exist prior to 2028.

Corporate information

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Senior VP, General Counsel, Chief Compliance Officer & Secretary J. Colella	CEO, President & Chairman B. Loy
Senior VP & CTO J. A. O'Neill	Senior VP & CFO L. LaGorga
VP, Controller & Chief Accounting Officer M. D. Sauer	
Board Members	
A. Saleki-Gerhardt	J. R. Anderson
B. Loy	R. Clark
J. F. Gentilcore	Y. M. Gordon-Kanouff
J. P. Lederer	
Domicile Delaware	
Auditor KPMG LLP - Klynveld Peat Marwick Goerdeler	
Founded 1966	
Employees 10,000	
Stockholders 1,080	

Entegris, Inc.

Quantitative Evaluations						Expanded Ratio Analysis					
Fair Value Rank	1	2	3	4	5		2022	2021	2020	2019	
	LOWEST				HIGHEST		Price/Sales	2.86	8.23	7.04	4.30
	Based on CFRA's proprietary quantitative model, stocks are ranked from most overvalued [1] to most undervalued [5].					Price/EBITDA	11.05	27.10	24.62	17.97	
						Price/Pretax Income	38.00	39.51	36.96	21.51	
Fair Value Calculation	USD 48.64	Analysis of the stock's current worth, based on CFRA's proprietary quantitative model suggests that ENTG is overvalued by USD 39.18 or 44.61%				P/E Ratio	17.58	40.28	37.83	25.95	
						Avg. Diluted Shares Outstg. [M]	143.15	136.57	136.27	136.57	
						Figures based on fiscal year-end price					
Volatility	LOW		AVERAGE		HIGH	Key Growth Rates and Averages					
Technical Evaluation	BEARISH	Since October, 2023, the technical indicators for ENTG have been BEARISH"				Past Growth Rate [%]					
Insider Activity	UNFAVORABLE	NEUTRAL		FAVORABLE		1 Year					
						3 Years					
						5 Years					
						Net Income					
						Sales					
						Ratio Analysis (Annual Avg.)					
						Net Margin [%]					
						% LT Debt to Capitalization					
						Return on Equity [%]					

Company Financials Fiscal year ending Dec 31										
Per Share Data [USD]	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
Tangible Book Value	-20.33	4.32	2.18	1.02	1.22	3.19	2.38	1.43	0.71	5.05
Free Cash Flow	-0.80	1.40	2.34	2.00	1.44	1.41	1.01	0.35	0.44	0.35
Earnings	1.46	3.00	2.16	1.87	1.69	0.59	0.68	0.57	0.06	0.53
Earnings [Normalized]	3.73	3.44	2.54	1.93	1.89	1.44	0.94	0.85	0.69	0.59
Dividends	0.40	0.34	0.32	0.31	0.28	0.14	N/A	N/A	N/A	N/A
Payout Ratio [%]	27.43	10.64	14.66	15.92	16.44	11.63	NM	NM	NM	NM
Prices: High	141.82	158.00	100.12	51.52	39.55	33.05	18.95	15.20	14.05	11.65
Prices: Low	61.75	93.50	38.12	27.39	23.39	17.65	10.37	11.90	10.20	8.75
P/E Ratio: High	38.00	45.90	39.40	26.70	20.90	23.00	20.20	17.90	20.40	19.70
P/E Ratio: Low	16.60	27.20	15.00	14.20	12.40	12.30	11.00	14.00	14.80	14.80
Income Statement Analysis [Million USD]										
Revenue	3,282	2,299	1,859	1,591	1,550	1,343	1,175	1,081	962.00	693.00
Operating Income	570.00	560.00	395.00	239.00	308.00	253.00	162.00	131.00	124.00	93.00
Depreciation + Amortization	279.00	138.00	137.00	141.00	127.00	102.00	100.00	102.00	84.00	39.00
Interest Expense	213.00	41.00	49.00	47.00	34.00	32.00	37.00	39.00	33.00	0.00
Pretax Income	247.00	479.00	354.00	318.00	254.00	185.00	120.00	90.00	-14.00	96.00
Effective Tax Rate	15.40	14.60	16.70	19.90	5.40	54.00	19.00	11.30	157.60	22.50
Net Income	209.00	409.00	295.00	255.00	241.00	85.00	97.00	80.00	8.00	75.00
Net Income [Normalized]	212.90	319.30	222.90	122.50	170.30	137.10	78.90	62.40	56.80	60.10
Balance Sheet and Other Financial Data [Million USD]										
Cash	594.00	403.00	581.00	352.00	482.00	625.00	406.00	352.00	394.00	384.00
Current Assets	2,335	1,313	1,234	932.00	1,029	1,058	800.00	709.00	766.00	612.00
Total Assets	10,139	3,192	2,918	2,516	2,318	1,976	1,700	1,647	1,762	875.30
Current Liabilities	762.00	379.00	303.00	264.00	270.00	291.00	262.00	176.00	263.00	98.00
Long Term Debt	5,633	937.00	1,086	932.00	935.00	574.00	485.00	606.00	667.00	N/A
Total Capital	9,103	2,722	2,515	2,156	1,951	1,667	1,484	1,459	1,515	757.00
Capital Expenditures	466.00	211.00	132.00	112.00	110.00	94.00	65.00	72.00	58.00	60.00
Cash from Operations	352.00	400.00	447.00	382.00	313.00	293.00	208.00	121.00	126.00	109.00
Current Ratio	3.06	3.47	4.08	3.53	3.82	3.63	3.06	4.04	2.92	6.27
% Long Term Debt of Capitalization	61.90	34.40	43.20	43.20	47.90	34.40	32.70	41.50	44.00	N/A
% Net Income of Revenue	6.40	17.80	15.90	16.00	15.50	6.30	8.30	7.40	0.80	10.70
% Return on Assets	5.35	11.46	9.10	6.19	8.98	8.60	6.05	4.79	5.89	6.92
% Return on Equity	8.50	26.50	23.20	23.40	24.00	9.00	11.40	10.40	1.00	10.30

Source: S&P Global Market Intelligence. Data may be preliminary or restated; before results of discontinued operations/special items. Per share data adjusted for stock dividends; EPS diluted. E-Estimated. NA-Not Available. NM-Not Meaningful. NR-Not Ranked. UR-Under Review.

Entegris, Inc.

Sub-Industry Outlook

Our fundamental outlook for the semiconductor equipment sub-industry for the next 12 months is positive. Foundries are sustaining elevated levels of investment (despite some reported delays in fab construction), while the memory market appears close to bottoming in what has been a rough '23, with softening demand and lower prices leading to meaningful production cuts. We see chip equipment sales rising 7% in '24 following a projected decline of 10% in '23. Political drivers remain a catalyst, with global tensions encouraging countries to localize their semiconductor supply chains. Major examples include the U.S. CHIPS Act, passed in '22, which provides \$52.7B in incentives, and the recently-passed European Chips Act, which provides \$47B. These tensions also create material risks for the industry, especially for equipment makers tied to the leading edge, as the U.S. attempts to restrict China from reaching high-volume capabilities in advanced nodes.

We estimate that memory spending will grow 55% in '24 following a projected decline of 45% in '23 (bottoming in Q3). NAND has been hit particularly hard in '23, with related investments down ~65%, while DRAM has held up relatively well (-19%) on increasing demand for high-bandwidth memory to support emerging AI applications (which are also driving leading-edge logic and back-end advanced packaging), a trend we think will continue in '24. We think previous consolidation in the memory space will allow customers to better sustain a healthy long-term industry landscape and create more predictable cyclical corrections moving forward. In addition, we think China's desire to grow its local chip manufacturing industry is supporting the demand for equipment, most visible in industrial end markets that rely on trailing-edge nodes.

Foundry spending through '24 will likely be driven by 5-nm and below investments. We estimate foundry spending will increase 5% in '24 following a flat-to-down year in '23 and sharp growth (>20%) in '22, driven by the continuation of aggressive expansion projects from industry

leader Taiwan Semiconductor [~\$32B in '24/'23 vs. \$35B in '22 and \$27B in '21], with continued investments from Intel [~\$21B in '24/'23 vs. ~\$25B in '22 and \$19B in '21] and Samsung planning to triple its foundry capacity by '26. We estimate foundries' share in total capex will be about 65%-85% over the next year, and industry sales will continue to be concentrated among these three companies, estimated to comprise ~60% of '24 sales.

We note that equipment providers exposed to the flat panel display industry should benefit from greater penetration for Organic LED (OLED) displays in the long term, offset by continued soft consumer demand in '24. We also see rising orders supported by the emergence of new products and improving demand for devices such as PCs and handsets following a tough '23.

For the most part, balance sheets at the largest equipment providers are healthy and could sustain pronounced cyclical downturns similar to those seen in the past, in our view. For '24, we expect free cash flow to increase modestly, as capex remains elevated and cash from operations improves, which will support greater share repurchases and higher dividend payments to maximize shareholder value, in our view.

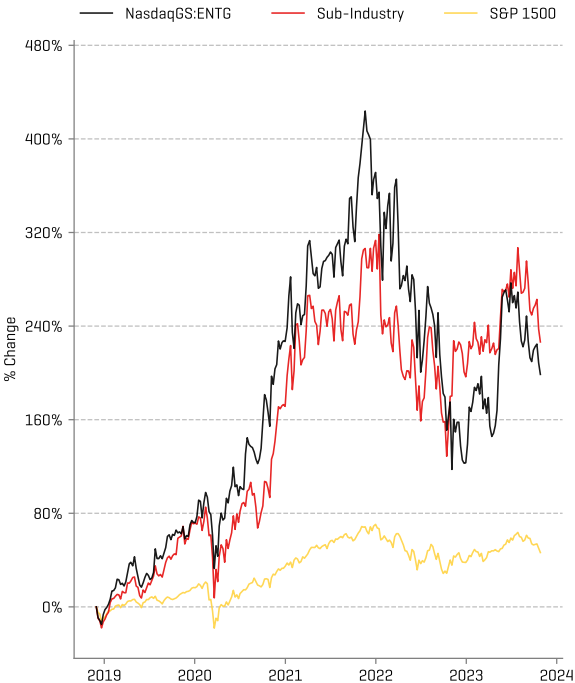
YTD through September 29, the S&P Semiconductor Equipment Index increased 19.7% vs. a 10.9% rise for the S&P 1500. In 2022, the S&P Semiconductor Equipment Index declined 28.2% vs. a contraction of 19.1% for the S&P 1500.

/ Brooks Idlet, CFA

Industry Performance

GICS Sector: Information Technology
Sub-Industry: Semiconductor Materials and Equipment

Based on S&P 1500 Indexes
Five-Year market price performance through Oct 28, 2023



NOTE: A sector chart appears when the sub-industry does not have sufficient historical index data.
All Sector & Sub-Industry information is based on the Global Industry Classification Standard (GICS).
Past performance is not an indication of future performance and should not be relied upon as such.
Source: CFRA, S&P Global Market Intelligence

Sub-Industry: Semiconductor Materials and Equipment Peer Group*: Semiconductor Materials and Equipment

Peer Group	Stock Symbol	Exchange	Currency	Recent Stock Price	Stk. Mkt. Cap. (M)	30-Day Price Chg. (%)	1-Year Price Chg. (%)	P/E Ratio	Fair Value Calc.	Yield (%)	Return on Equity (%)	LTD to Cap (%)
Entegris, Inc.	ENTG	NasdaqGS	USD	87.53	13,139.0	-1.4	9.0	29.0	48.64	0.5	3.5	61.6
Amkor Technology, Inc.	AMKR	NasdaqGS	USD	21.23	5,218.0	-4.3	5.8	9.0	25.05	1.4	16.6	19.2
Axcelis Technologies, Inc.	ACLS	NasdaqGS	USD	138.74	4,553.0	-9.9	136.6	22.0	N/A	N/A	31.0	N/A
Enphase Energy, Inc.	ENPH	NasdaqGM	USD	96.18	13,074.0	-19.3	-67.0	17.0	N/A	N/A	80.4	52.4
FormFactor, Inc.	FORM	NasdaqGS	USD	31.70	2,462.0	-4.0	16.8	NM	N/A	N/A	-0.9	1.6
Kulicke and Soffa Industries, Inc.	KLIC	NasdaqGS	USD	42.19	2,383.0	-11.9	0.1	25.0	N/A	1.8	8.3	N/A
MKS Instruments, Inc.	MKSI	NasdaqGS	USD	68.02	4,548.0	-18.2	-16.1	10.0	N/A	1.3	-62.3	63.2
Nova Ltd.	NVMI	NasdaqGS	USD	91.25	2,617.0	-15.5	23.8	21.0	N/A	N/A	22.9	N/A
Onto Innovation Inc.	ONTO	NYSE	USD	113.93	5,590.0	-6.7	70.0	32.0	N/A	N/A	10.9	N/A
SolarEdge Technologies, Inc.	SEDG	NasdaqGS	USD	83.41	4,717.0	-34.5	-63.0	9.0	78.93	N/A	13.3	19.2
Teradyne, Inc.	TER	NasdaqGS	USD	85.12	13,110.0	-11.2	3.5	28.0	N/A	0.5	21.5	N/A

*For Peer Groups with more than 10 companies or stocks, selection of issues is based on market capitalization.
NA-Not Available; NM-Not Meaningful.
Note: Peers are selected based on Global Industry Classification Standards and market capitalization. The peer group list includes companies with similar characteristics, but may not include all the companies within the same industry and/or that engage in the same line of business.

Entegris, Inc.**Analyst Research Notes and other Company News****August 03, 2023**

05:33 PM ET... CFRA Maintains Buy Recommendation on Shares of Entegris, Inc. [ENTG 102.74****]:

We raise our 12-month target by \$19 to \$125, 33.7x our 2024 EPS view, a slight premium to ENTG's three-year forward P/E average of 32.1x. We raise our 2023 EPS estimate to \$2.64 from \$2.57 and 2024's to \$3.71 from \$3.53. ENTG printed Q2 EPS of \$0.66 vs. \$1.00 a year ago, \$0.09 above consensus. Revenue contracted 2% sequentially, but was up 30% Y/Y due to the acquisition of CMC. The Microcontamination Control segment saw 5% Q/Q growth driven by liquid filtration and gas purification solutions. However, weakness was seen in Advanced Materials Handling [-13% Q/Q] on weakness in products tied to WFE like FOUPs. Advanced Planarization Solutions [-4% Q/Q] also came in light as lower volumes were observed in most product lines. With elevated debt of \$5.6 billion [gross leverage ratio of 5.2x], we are encouraged with ENTG's priority to pay down debt and note ENTG is expected to receive at least \$1 billion in proceeds from the sale of businesses this year; positively, ENTG is using all the proceeds to pay down debt. / Alexander Yokum, CFA

May 11, 2023

09:58 AM ET... CFRA Maintains Buy Recommendation on Shares of Entegris, Inc. [ENTG 93.00****]:

We keep our 12-month target at \$106, 30x our 2024 EPS view, below historical. We cut our 2023 EPS estimate to \$2.57 from \$3.20 and 2024's to \$3.53 from \$4.25. ENTG prints Q1 EPS of \$0.65 vs. \$1.06, above our expectations. ENTG revenue increased 42% Y/Y [-2.5% Q/Q] due to closing the acquisition of CMC for \$922M. Advanced Materials Handling sales increased 10% Y/Y [+2%] to \$219M, due to wafer/fluid handling solutions. Microcontamination Control rose 1% [-5%] to \$269M, a sequential decline due to capex-driven solutions in WFE slowdown. This is partially offset by a 16% decline [-1%] in Advanced Planarization Solutions to \$250M, due to lower CMP consumable sales [excluding SIC slurries]. Specialty Chemicals/Engineered Materials sales fell 6% [-3%] due to the sale of the QED business mid-Q1 [for around \$153M]. We expect wafer starts to bottom in Q2. ENTG also entered into an agreement to sell the Electronic Chemicals business for \$700M to pay down debt [net debt/EBITDA 6.1x], which we view favorably. / Keven Young

February 14, 2023

11:46 AM ET... CFRA Maintains Buy Recommendation on Shares of Entegris, Inc. [ENTG 82.51****]:

We raise our 12-month target \$14 to \$106, 25x our 2024 EPS view, above semi-cap peers due to its greater exposure to recurring revenue and track record of outperforming the WFE. We lower our 2023 EPS to \$3.20 from \$3.67 and keep 2024 at \$4.25. ENTG had a Q4 adj-EPS of \$0.83 vs. \$0.96 [\$0.78 consensus]. ENTG realized 48.9% revenue growth in Q4 Y/Y [\$946M] due to the closed acquisition of CMC Materials. Microcontamination Control [MC] sales grew 10% due to gas purification and liquid filtration, and Advanced Materials Handling [AMH] revenue rose 8.2% led by wafer/fluid handling and liquid packaging solutions. Specialty Chemicals and Engineered Materials [SCEM] was flat Y/Y due to export controls. Advanced Planarization Solutions [APS] revenue fell about 11% in Q4 due to the softening semiconductor market. We are encouraged by the guidance for a bottom in end-market demand in Q2, in line with our view. We see ENTG outperforming WFE by 500 bps in 2023 on MC/AMH, but we expect SCEM to lag from exposure to memory. / Keven Young

December 29, 2022

11:43 AM ET... CFRA Downgrades Opinion of Shares of Entegris, Inc. to Buy from Strong Buy [ENTG 65.39****]:

We are downgrading ENTG shares to Buy from Strong Buy due to a dampened outlook for wafer starts in the first half of 2023 with a pronounced decline in memory chip manufacturing facility utilization now more than offsetting trailing edge strength, which is largely tied to automotive and industrial applications. We see ENTG as the best-in-class specialty chemical and material integrity solutions provider for semiconductor, microelectronics, and life sciences market, but are muted on what investors are willing to pay for the name in short term due to weaker end-market demand and leveraged balance sheet following the acquisition of CMC Materials. We lower our 12-month target price by \$35 to \$92 based on a P/E of 25x our 2023 EPS view, above peers due to recurring revenue exposure and a track record of outperforming WFE. We cut our 2022 EPS estimate to \$3.69 from \$3.75, lower our 2023 EPS view to \$3.67 from \$4.26, while also cutting our 2024 EPS

estimate to \$4.25 from \$4.85. / Keven Young

November 02, 2022

04:28 PM ET... CFRA Maintains Strong Buy Recommendation on Shares of Entegris, Inc. [ENTG 63.51****]:

We lower our 12-month target by \$28 to \$127 based on P/E of 30x our 2023 EPS view, above peers due to recurring revenue exposure and track record of outperforming WFE. We cut our 2022 EPS view to \$3.75 from \$4.14, lower our 2023 EPS estimate to \$4.26 from \$4.85, and initiate on 2024 EPS view at \$4.85. ENTG prints Q3 adj-EPS of \$0.85 vs \$0.92, missing the \$1.00 consensus. Microcontamination Control [MC] revenue increased 24% to \$281M with strength across major product lines. Specialty Chemicals & Engineered Materials [SCEM] sales rose 13.6% to \$224M driven by surface preparation solutions and advanced deposition materials. Advanced Materials handling [AMH] revenue grew 13% led by wafer and fluid handling solutions, offset by Armus bag sales decline. Advanced Planarization Solutions [APS] sales increased 8.9% to \$294M primarily driven by SMP consumable products. We expect Q4 revenue growth to be driven by MC's liquid and gas filtration and gas purification, along with SCEM's advanced deposition materials. / Keven Young

August 02, 2022

10:05 AM ET... CFRA Maintains Strong Buy Recommendation on Shares of Entegris, Inc. [ENTG 106.40****]:

We keep our 12-month target at \$155 based on a P/E of 32x our 2023 EPS view, above semi-cap peers at ~20x warranted by ENTG's growing exposure to recurring revenue streams. We keep our 2022 EPS estimate of \$4.14 and raise our 2023 EPS view to \$4.85 from \$4.60. ENTG prints Q2 adj-EPS of \$1.00 vs \$0.85, missing the \$1.05 consensus, with a 290 bps contraction in gross margin Y/Y due to an FX headwind largely related to the USD/JPY. ENTG realized 21% Y/Y revenue growth in Q2 to \$692M led by AMH sales which rose ~30% [\$224M] due to products for high level fab investments [e.g., wafer/fluid handling & measurement solutions], followed by ~\$21% growth in MC [\$274M] due to gas/liquid filtration & purification products, alongside SCEM which grew revenue 15% [\$208M] due to advanced deposition materials & preparation solutions. We note this quarter reporting was entirely legacy ENTG [not CCMP]; we are encouraged by the integration of CCMP and expect it to drive growth and reduce cyclicality for the combined entity. / Keven Young

July 06, 2022

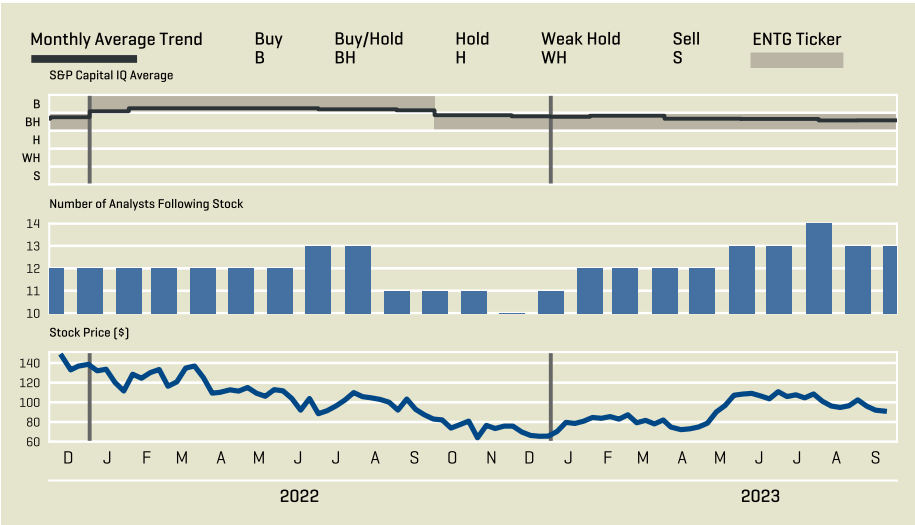
11:01 AM ET... CFRA Maintains Strong Buy Recommendation on Shares of Entegris, Inc. [ENTG 91.05****]:

ENTG closed its acquisition of CMC Materials [CCMP] today, creating a global leader in electronic materials. We think that the complementary portfolio of CCMP adds value to ENTG's existing unit-driven platform, allowing the combined entity to service customers demanding process challenges and to better support their ambitious technology roadmaps. The total purchase price at close was approximately \$5.7B, with \$3.8B in cash paid to CCMP shareholders, alongside 12.9M ENTG shares, and ~\$900M in debt retired along with \$200M in cash acquired. We expect the acquisition to accretive to non-GAAP EPS within on year, expected \$75M in run-rate cost synergies & \$40M in CapEx synergies within 12-18 months. The acquisition also comes with a 4th reporting segment for ENTG, "Advanced Planarization Solutions [APS] Division", which will include suite of CMP solutions, CMP Slurries, CMP Pads, Electronic Chemicals businesses, CMP Cleans, Pad Conditioners, and more. We keep our 12-month target of \$155, 33.7x our '23 EPS view. / Keven Young

Note: Research notes reflect CFRA's published opinions and analysis on the stock at the time the note was published. The note reflects the views of the equity analyst as of the date and time indicated in the note, and may not reflect CFRA's current view on the company.

Entegris, Inc.

Analysts Recommendations



	No. of Recommendations	% of Total	1 Mo. Prior	3 Mos. Prior
Buy	5	38	5	5
Buy/Hold	4	31	4	5
Hold	4	31	4	3
Weak hold	0	0	0	0
Sell	0	0	0	0
No Opinion	0	0	0	0
Total	13	100	13	13

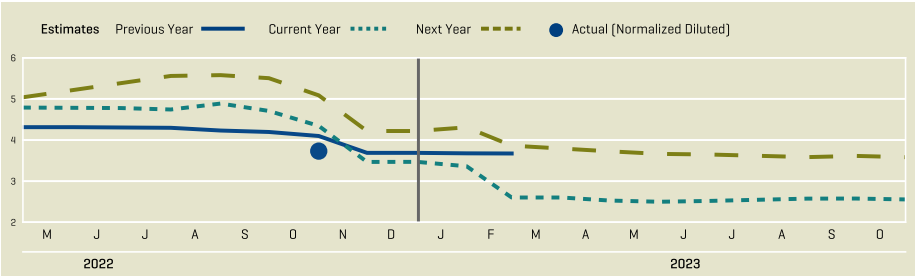
Wall Street Consensus Opinion

Buy/Hold

Wall Street Consensus vs. Performance

For fiscal year 2023, analysts estimate that ENTG will earn USD 2.55. For fiscal year 2024, analysts estimate that ENTG's earnings per share will grow by 40.26% to USD 3.58.

Wall Street Consensus Estimates



Fiscal Year	Avg Est.	High Est.	Low Est.	# of Est.	Est. P/E
2024	3.58	4.18	3.15	12	24.43
2023	2.55	2.69	2.48	12	34.27
2024 vs. 2023	▲ 40%	▲ 55%	▲ 27%	N/A%	▼ -29%
Q3'24	0.94	1.20	0.79	4	92.80
Q3'23	0.60	0.62	0.59	12	144.70
Q3'24 vs. Q3'23	▲ 56%	▲ 94%	▲ 34%	▼ -67%	▼ -36%

Forecasts are not reliable indicator of future performance.
Note: A company's earnings outlook plays a major part in any investment decision. S&P Global Market Intelligence organizes the earnings estimates of over 2,300 Wall Street analysts, and provides their consensus of earnings over the next two years, as well as how those earnings estimates have changed over time. Note that the information provided in relation to consensus estimates is not intended to predict actual results and should not be taken as a reliable indicator of future performance.
Note: For all tables, graphs and charts in this report that do not cite any reference or source, the source is S&P Global Market Intelligence.

Entegris, Inc.

Glossary

STARS

Since January 1, 1987, CFRA Equity and Fund Research Services, and its predecessor S&P Capital IQ Equity Research has ranked a universe of U.S. common stocks, ADRs [American Depositary Receipts], and ADSs [American Depositary Shares] based on a given equity's potential for future performance. Similarly, we have ranked Asian and European equities since June 30, 2002. Under proprietary STARS [Stock Appreciation Ranking System], equity analysts rank equities according to their individual forecast of an equity's future total return potential versus the expected total return of a relevant benchmark [e.g., a regional index (MSCI AC Asia Pacific Index, MSCI AC Europe Index or S&P 500® Index)], based on a 12-month time horizon. STARS was designed to help investors looking to put their investment decisions in perspective. Data used to assist in determining the STARS ranking may be the result of the analyst's own models as well as internal proprietary models resulting from dynamic data inputs.

S&P Global Market Intelligence's Quality Ranking

[also known as **S&P Capital IQ Earnings & Dividend Rankings**] - Growth and S&P Capital IQ Earnings & Dividend Rankings stability of earnings and dividends are deemed key elements in establishing S&P Global Market Intelligence's earnings and dividend rankings for common stocks, which are designed to capsize the nature of this record in a single symbol. It should be noted, however, that the process also takes into consideration certain adjustments and modifications deemed desirable in establishing such rankings. The final score for each stock is measured against a scoring matrix determined by analysis of the scores of a large and representative sample of stocks. The range of scores in the array of this sample has been aligned with the following ladder of rankings:

A+ Highest	B Below Average
A High	B- Lower
A Above	C Lowest
B+ Average	D In Reorganization
NC Not Ranked	

EPS Estimates

CFRA's earnings per share (EPS) estimates reflect analyst projections of future EPS from continuing operations, and generally exclude various items that are viewed as special, non-recurring, or extraordinary. Also, EPS estimates reflect either forecasts of equity analysts; or, the consensus [average] EPS estimate, which are independently compiled by S&P Global Market Intelligence, a data provider to CFRA. Among the items typically excluded from EPS estimates are asset sale gains; impairment, restructuring or merger-related charges; legal and insurance settlements; in process research and development expenses; gains or losses on the extinguishment of debt; the cumulative effect of accounting changes; and earnings related to operations that have been classified by the company as discontinued. The inclusion of some items, such as stock option expense and recurring types of other charges, may vary, and depend on such factors as industry practice, analyst judgment, and the extent to which some types of data is disclosed by companies.

12-Month Target Price

The equity analyst's projection of the market price a given security will command 12 months hence, based on a combination of intrinsic, relative, and private market valuation metrics, including Fair Value.

Abbreviations Used in Equity Research Reports

- CAGR - Compound Annual Growth Rate
- CAPEX - Capital Expenditures
- CY - Calendar Year
- DCF - Discounted Cash Flow
- DDM - Dividend Discount Model
- EBIT - Earnings Before Interest and Taxes
- EBITDA - Earnings Before Interest, Taxes, Depreciation & Amortization
- EPS - Earnings Per Share
- EV - Enterprise Value
- FCF - Free Cash Flow
- FFO - Funds From Operations
- FY - Fiscal Year
- P/E - Price/Earnings
- P/NAV - Price to Net Asset Value
- PEG Ratio - P/E-to-Growth Ratio
- PV - Present Value
- R&D - Research & Development
- ROCE - Return on Capital Employed
- ROE Return on Equity
- ROI - Return on Investment
- ROIC - Return on Invested Capital
- ROA - Return on Assets
- SG&A - Selling, General & Administrative Expenses
- SOTP - Sum-of-The-Parts
- WACC - Weighted Average Cost of Capital

Dividends on American Depositary Receipts (ADRs) and American Depositary Shares (ADSs) are net of taxes [paid in the country of origin].

Qualitative Risk Assessment

Reflects an equity analyst's view of a given company's operational risk, or the risk of a firm's ability to continue as an ongoing concern. The Qualitative Risk Assessment is a relative ranking to the U.S. STARS universe, and should be reflective of risk factors related to a company's operations, as opposed to risk and volatility measures associated with share prices. For an ETF this reflects on a capitalization-weighted basis, the average qualitative risk assessment assigned to holdings of the fund.

STARS Ranking system and definition:

★★★★★ 5-STARS (Strong Buy):

Total return is expected to outperform the total return of a relevant benchmark, by a notable margin over the coming 12 months, with shares rising in price on an absolute basis.

★★★★★ 4-STARS (Buy):

Total return is expected to outperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 3-STARS (Hold):

Total return is expected to closely approximate the total return of a relevant benchmark over the coming 12 months.

★★★★★ 2-STARS (Sell):

Total return is expected to underperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 1-STAR (Strong Sell):

Total return is expected to underperform the total return of a relevant benchmark by a notable margin over the coming 12 months, with shares falling in price on an absolute basis.

Relevant benchmarks:

In North America, the relevant benchmark is the S&P 500 Index, in Europe and in Asia, the relevant benchmarks are the MSCI AC Europe Index and the MSCI AC Asia Pacific Index, respectively.



Entegris, Inc.

Disclosures

Stocks are ranked in accordance with the following ranking methodologies:

STARS Stock Reports:

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Quantitative Stock Reports:

Quantitative rankings are determined by ranking a universe of common stocks based on 5 measures or model categories: Valuation, Quality, Growth, Street Sentiment, and Price Momentum. In the U.S., a sixth sub-category for Financial Health will also be displayed. Percentile scores are used to compare each company to all other companies in the same universe for each model category. The five [six] model category scores are then weighted and rolled up into a single percentile ranking for that company. For reports containing quantitative rankings refer to the Glossary section seof the report for detailed methodology and the definition of Quantitative rankings.

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Global STARS Distribution as of September 30, 2023

Ranking	North America	Europe	Asia	Global
Buy	36.8%	34.5%	41.6%	37.4%
Hold	53.8%	51.2%	50.3%	52.6%
Sell	9.4%	14.3%	8.1%	10.1%
Total	100.0%	100.0%	100.0%	100.0%

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Entegris, Inc.

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